PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)
(Estd. under the Panjab University Act VII of 1947—enacted by the Govt. of India)

FACULTY OF ARTS

SYLLABI

FOR

ADVANCED DIPLOMA IN LIBRARY AUTOMATION AND NETWORKING

EXAMINATIONS 2023-24, 2024-25, 2025-26

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SYLLABUS

ADVANCED DIPLOMA IN LIBRARY AUTOMATION AND NETWORKING (ADLAN)
(SEMESTER SYSTEM)

Paper Code	Title of the Paper	Marks			
		INTERNAL ASSESSMENT*	EXTERNAL EXAM	TOTAL	EXAM HOURS
1st Semester					
LAN- 01	Basics of Computers and Network Technology	20	80	100	3
LAN- 02	Library Automation	20	80	100	3
LAN- 03	Computer Practical-1	20	80	100	3
2nd Semester					
LAN- 04	Web-based Resources	20	80	100	3
LAN- 05	Digital Library and Content Management	20	80	100	3
LAN- 06	Computer Practical-2	20	80	100	3

* For the purpose of Internal Assessment in all papers, two assignments will be given (10 marks each), which will be compulsory.

1st Semester

Paper Code: LAN-01

Paper Title: Basics of Computers and Network Technology

Instructions for the Paper-Setters / Examiners:
The Paper carries 80 marks. Duration of the paper is 3 hours. There should be 9 questions in all, out of which the candidate be asked to attempt 5 questions, selecting one question from each unit. The first question shall be short answer type, containing 15 short questions spread over the whole syllabus, to be answered in about 25-30 words each. The candidate is required to attempt any 10 short answer type questions. It shall carry 20 marks @ 2 marks for each short question and shall be compulsory question. Rest of the paper shall contain 4 units. Each unit shall have 2 questions and the candidates shall be given internal choice i.e. the candidate shall attempt one question from each unit. The questions should be evenly distributed within the units. In no case a question should be asked from outside the syllabus. The question paper should be strictly according to the instructions mentioned above.

Objectives:
To provide basic knowledge of Computers and networking.

Unit-1

Computer Fundamentals
Computers and their Components: Definition, characteristics, generations, functional units, input devices, output devices, secondary storage devices, central processing unit.
Storage Media: RAM, ROM, Flash Memory, Optical storage Devices, Cloudand Virtual storage.

Unit-II

Software
Software: Types of software.
Operating System: Functions, types- Windows, Linux, Ubuntu, Android and ios
Application Software: Using Windows, MS- Office (Word, Excel, and PowerPoint)

Unit-III

Introduction to Networks

Types of Networks: Uses and advantages of networking, types of networks, network topologies, Network components. Hardware: Network Adaptors, Modems, Switch, Hub etc., Transmission Media.

Network Technology: OSI model, Virtual Private Network (VPN), Internet TCP/IP Models. Data Networks: Data transmission and signaling, ISDN, DSL, ATM.

Network Software: Network architecture and role of network software, Client Server architecture, Network operating systems, Network management system and protocols.

Unit-IV

Internet Technology

Internet: Definitions, History, Organizations associated with Internet. Internet Architecture Web 2.0 technologies and services: RSS, Podcasts and Vodcasts, Blogs, Wikis, Social Networks. Semantic Web, AI tools and their applications.

Essential Readings

1. Alkhatib, G. I. (Ed.). (2012). *Network and Communication Technology Innovations for Web and IT Advancement*. IGI Global.

2. Balasubramanian, P. (2011). *Library information and communication technology*. New Delhi: Deep & Deep.

3. Clark, Tom (2018). *Storage virtualizations technologies for simplifying data storage and managements.*Bengaluru: Person India.

4. Hardeep Singh, Kakkar, S. K., & Sharma, A. (2011). *A book of fundamental of information technology*. Amritsar: Lakhanpal Pub.

5. Stallings. & Brown, L. (2019). *Computer security: Principles and practice.* Boston: Pearson.

6. Sinha, P. K., & Sinha, P. (2022). *Computer fundamentals* (8th ed.). New Delhi: BPB.

7. Tanenbaum, A.S., &Feamster, N. (2019). *Computer networks.* Boston: Pearson

8. Williams, B.K., & Sawyer, S.C. (2015). *Using information technology: A practical introduction to computers & communications.* New York: McGraw Hill.

Further Readings

1. Balagurusamy, E. (2010). *Fundamentals of Computers*. New Delhi: Tata McGraw Hill

2. Sharma, S. K. (2007). *Information technology and library services*. New Delhi: Shree Publishers and Distributors.

3. Venkata, P. R. (2004). *Information technology applications in libraries*. Delhi: EssEss Publications.

Paper Code: LAN-02

Paper Title: Library Automation

Instructions for the Paper-Setters / Examiners:

The Paper carries 80 marks. Duration of the paper is 3 hours. There should be 9 questions in all, out of which the candidate be asked to attempt 5 questions, selecting one question from each unit. The first question shall be short answer type, containing 15 short questions spread over the whole syllabus, to be answered in about 25-30 words each. The candidate is required to attempt any 10 short answer type questions. It shall carry 20 marks @ 2 marks for each short question and shall be compulsory question. Rest of the paper shall contain 4 units. Each unit shall have 2 questions and the candidates shall be given internal choice i.e., the candidate shall attempt one question from each unit. The questions should be evenly distributed within the units. In no case a question should be asked from outside the syllabus. The question paper should be strictly according to the instructions mentioned above.

Objectives:
To provide information about latest concepts of Library Automation Software.

Unit -1

Library automation

- Library Automation: Need, purpose, objectives and its area of application.
- Library Automation Process: Planning, implementation and evaluation. Approaches to Library Automation.
- Advantages and barriers of Library Automation.

Unit-II

Bibliographic Formats and Standards

- Need, Standards and various options in retrospective conversion
- Formats and Standards: MARC, UNIMARC, CCF, FRBR and RDA

Unit-III

ILMS

- ILMS: Concept, features and types
- BiblioteQ: Introduction, features and installation
- KOHA: Introduction, features and installation
- eGranthalaya: Introduction, features and installation

Unit-IV

Inventory Management in Libraries: toolsand techniques

- Barcode technology: Need, purpose and implementation
- RFID technology: Need, purpose and implementation
- QR code technology: Need, purpose and implementation

Essential Readings:

1. Aswal, R. S. (2006). *Library automation for 21st Century*. Delhi: EssEss Publication.
2. Balasubramanian, P. (2011). *Library automation & networking*. New Delhi: Deep & Deep.
3. Bilal, D. (2014). *Library automation: Core concepts and practical systems analysis.* California: Libraries Unlimited.
4. Clayton, Marlene (2018). *Managing library automation*. London: Routledge
5. Kulkarni, S., &Shewale, N. (2017). *Enhancing library services using open-source software Koha: Prototype model of centralized library system of Sinhgad Institute's Higher Education libraries.*S.l.: Lambert Academic Publishing
6. Nagia, R. (2012). *Making of software packages for library automation*. New Delhi: Cyber Tech. Pub.

Further Readings:

1. Singh, P. K. (2005). *Library automation*. New Delhi: Shree.

2. Theresa, T. B., Ratna, K. C., & Rai, B. A. (2011). *Information technology and library automation.* New Delhi: Commonwealth Publisher.

Paper Code: LAN-03

Paper Title: Computer Practical-1

Instructions for the Paper-Setters / Examiners

The Paper carries 80 marks. Duration of the paper is 3 hours. There will be 2 parts, out of which Part- I is compulsory based on **viva-voce** of 20 marks. Part - II contains 4 units. Each unit shall have 2 questions and the candidates shall be given internal choice i.e., the candidate shall attempt one question from each unit. The questions should be evenly distributed within the units. In no case a question should be asked from outside the syllabus. The question paper should be strictly according to the instructions mentioned above.

Objectives:

To develop computer skills and professional competencies to solve problems of libraries and information centers through practical knowledge of computers.

Unit-I

MS- Windows & MS- Office

- MS WINDOWS: File management.
- MS-OFFICE: MS WORD, MS EXCEL and MS POWERPOINT**.** MS ACCESS.

Unit-II

Library Automation

- Library Automation: Installation & Working with BiblioteQ, KOHA and eGranthalaya
- Dewey Cutter Program (OCLC).

Unit-III

Library 2.0 Tools

- RSS feeds. Social Bookmarking and Tagging. Blogs, Wikis

Unit-IV

Networking and AI Tools

- Resource Sharing: File, Folder and printer sharing
- Networking tools and services
- AI tools: Grammerly, Scholarcy, Chatbots

Essential Readings

1. Bilal, D. (2014). Library automation: Core concepts and practical systems analysis. California: Libraries Unlimited.

2. Gurvinder Singh, &Rachhpal Singh. (2007). *A text book of windows-based computer courses* (9th ed.). Ludhiana: Kalyani Publishers.

3. Norton, P. (2005). *Introduction to computers* (6th ed.). New Delhi: Tata McGraw Hill.

4. Sinha, P. K., & Sinha, P. (2022). *Computer fundamentals* (8th ed.). New Delhi: BPB.

Further Readings

1. Bradley, P. (2007). *How to use web 2.0 in your library*. London: Facet.
2. Tanenbaum, A. S. (2009). *Computer networks* (4th ed.). New Delhi: Pearson.

2nd Semester

Paper Code: LAN-04

Paper Title: Web based resources

Instructions for the Paper-Setters / Examiners:

The Paper carries 80 marks. Duration of the paper is 3 hours. There should be 9 questions in all, out of which the candidate be asked to attempt 5 questions, selecting one question from each unit. The first question shall be short answer type, containing 15 short questions spread over the whole syllabus, to be answered in about 25-30 words each. The candidate is required to attempt any 10 short answer type questions. It shall carry 20 marks @ 2 marks for each short question and shall be compulsory question. Rest of the paper shall contain 4 units. Each unit shall have 2 questions and the candidates shall be given internal choice i.e., the candidate shall attempt one question from each unit. The questions should be evenly distributed within the units. In no case a question should be asked from outside the syllabus. The question paper should be strictly according to the instructions mentioned above.

Objectives:

To provide basic knowledge of web-based resources.

Unit-I

Web based resources

- Concept, meaning and types (Portals, gateways, databases, subject guides, aggregators)
- Criteria for evaluation of web-based resources

Unit-II

Discipline based web resources

- Science & Technology
- Social Sciences
- Arts and Humanities
- Health Science

Unit-III

Databases and Repositories

- Databases: Web of Science, Scopus, ERIC and LISTA
- Repositories: Institutional, Subject, ETD and Internet

Unit-IV

Academic Networking Platforms

- Research Gate
- Google Scholar
- Academia.edu
- Researcher App

Essential Readings

1. Botyriute, Kristina (2018). Access to Online Resources: A Guide for the Modern Librarian. https://link.springer.com/content/pdf/10.1007/978-3-319-73990-8.pdf

2. Callicott, B. B., Scherer, D., &Wesolek, A. (Eds.). (2016). *Making institutional repositories work*. Purdue University Press.

3. Christopher, D. M., Prabhakar, R., & Hinrich, S. (2008). *Introduction to information retrieval*. Cambridge University Press.
4. Harris, R. (2020). *Evaluating internet research sources*. https://www.virtualsalt.com/evaluating-internet-research-sources
5. Veletsianos, G. (2016). *Social media in academia: Networked scholars*. Routledge, Taylor & Francis Group.

Further Readings

1. Alkhatib, G. (2013). Network and communication technology innovations for web and IT advancement. Hershey, Pennsylvania: IGI Global.
2. Simons, N., & Richardson, J. (2013). *New content in digital repositories: The changing research landscape*. Chandos Pub.
3. Warekar, A. (2010). *Library science research and database books* Cyber Tech Publications.

Paper Code: LAN-05

Paper Title: Digital Library and Content Management

Instructions for the Paper-Setters / Examiners:

The Paper carries 80 marks. Duration of the paper is 3 hours. There should be 9 questions in all, out of which the candidate be asked to attempt 5 questions, selecting one question from each unit. The first question shall be short answer type, containing 15 short questions spread over the whole syllabus, to be answered in about 25-30 words each. The candidate is required to attempt any 10 short answer type questions. It shall carry 20 marks @ 2 marks for each short question and shall be compulsory question. Rest of the paper shall contain 4 units. Each unit shall have 2 questions and the candidates shall be given internal choice i.e., the candidate shall attempt one question from each unit. The questions should be evenly distributed within the units. In no case a question should be asked from outside the syllabus. The question paper should be strictly according to the instructions mentioned above.

Objectives:
To make students conversant with the Digital Library and Content Management.

Unit-I

Digital Library

- Digital Library: Definition, Objectives, Advantages, Components and Technical Infrastructure. Digitization steps, use of Scanner and OCR in digitization.
- Digital Preservation.
- IPR issues in digital library.

Unit-II

Digital Library Software and Metadata

- Metadata: Concept, types, Dublin Core Metadata Element Sets (DCMES). Digital Library Architecture
- Digital Library Software: Basic features of DSpace and Greenstone Digital Library (GSDL) Features and significance of Traditional Knowledge Digital Library (TKDL) and NDLI.

Unit-III

Content Analysis and Organisation:

- Content Analysis: Concept, uses and process.
- Qualitative & Quantitative aspects of Content Analysis.
- Techniques of Content Organisation: Classification and cataloguing of web documents, Subject Headings, Z39.50, SRU/SRW, Ontology, Folksonomy and DOI

Unit-IV

Content Management Systems (CMS)

- Content Management Systems:Concepts, features, types advantages.
- Features of Joomla, Drupal and Wordpress.
- Document Management System: Concepts and components.

Essential Readings

1. Balasubramanian, P. (2013). *E-learning for library professionals*. New Delhi: Regal.
2. Dahl, M., Banerjee, K., &Spalti, M. (2006). *Digital libraries: Integrating content and systems*. Oxford: Chandos.
3. Pathak, M. (2014). *An introduction to intellectual property rights*. New Delhi: New India Publishing Agency.
4. Verma, K. (2007). *Digital library preservation strategies*. New Delhi: Akansha.

Further Readings

1. Arms, W. Y. (2005). *Digital libraries*. New Delhi: Ane Books.
2. Hackos, J. (2002). *Content management for dynamic delivery*. New York: John Wiley.
3. Haynes, D. (2004). *Metadata for information management and retrieval*. London: Facet Pub.

Paper Code: LAN-06

Paper Title: Computer Practical-2

Instructions for the Paper-Setters / Examiners

The Paper carries 80 marks. Duration of the paper is 3 hours. There will be 2 parts, out of which Part- I is compulsory based on **viva-voce** of 20 marks. Part - II contains 4 units. Each unit shall have 2 questions and the candidates shall be given internal choice i.e., the candidate shall attempt one question from each

unit. The questions should be evenly distributed within the units. In no case a question should be asked from outside the syllabus. The question paper should be strictly according to the instructions mentioned above.

Objectives:

To develop computer skills and professional competencies to solve problems of libraries and information centers through practical knowledge of computers.

Unit-I

Search Strategies: Search Engines and Databases

- Framing search strategies for basic and advanced search for general search engine Google. Metasearch Engine: PolyMeta and Dogpile
- Effective use of Academic Search Engines like Google Scholar & BASE.
- WEB OPACs: Basic and advanced search interface of OPACs and Union Catalogues (PU Web OPAC, IndCat, WorldCat).

Unit-II

Search Techniques

- Searching ETDs, Abstract and Citation Databases: Shodhganga, Scopus and LISTA.
- Searching Open Access literature: DOAJ, OpenDOAR.
- Federated Searching: Tool, Techniques and Services.

Unit- III

Library Website and Reference Management

- Creating Library Website using Weebly and Google Sites.
- Reference Management Software: Mendeley.

Unit- IV

Digital Library

- Building Digital Library using Greenstone Digital Library Software (GSDL) and D-Space: a step-by-step approach including Metadata creation.

Essential Readings

1. Brown, C.C. & Bell, S.S. (2018). Librarian's guide to online searching: Cultivating database skills for research and instruction (5th ed.). California: Libraries Unlimited
2. Norton. (2017). Introduction to Computers. New Delhi: McGraw Hill Education.
3. Sinha, P. K., & Sinha, P. (2022). *Computer fundamentals* (6th ed.). New Delhi: BPB.
4. Tanenbaum, A. S. (2009). *Computer networks* (4th ed.). New Delhi: Pearson.
5. http://koha-community.org/
6. http://www.greenstone.org/

Further Readings

1. Broadbent, B. (2002). *ABCs of e-learning: Reaping the benefits and avoiding the pitfalls*. Jossey-Bass.
2. Gurvinder Singh, &Rachhpal Singh. (2007). A text book of windows-based computer courses (9th ed.). Ludhiana: Kalyani Publishers.